

Samuel Leblanc

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Education

Queen's University 2025 - 2027

M.Sc. in Mathematics

Research project: Cluster algebras, generalized cluster categories, and representation theory

Supervisors: [Ivan Dimitrov](#), [Charles Paquette](#), and [David Wehlau](#)

Université de Sherbrooke 2022 - 2025

B.Sc. in Mathematics

Mention of Excellence – Faculty Dean's List, Winter 2025

Research Interests

Representation theory of (finite-dimensional and associative) algebras, cluster algebras, homological algebra.

Papers

- Brüstle, T., Desrochers, J., Leblanc S., *Generalized Rank via Minimal Subposet*. [arXiv:2510.10837](#)  (2025)
- Leblanc, S., Rasolomanana, A., Armenta, M., *Hidden Activations Are Not Enough: A General Approach to Neural Network Predictions*. [arXiv:2409.13163](#)  (2024)
- Armenta, M., Leblanc, S., *Batalin-Vilkovisky structure on Hochschild cohomology with coefficients in the dual algebra*. [arXiv:1810.13023](#)  (2018) (I joined as coauthor in 2025; revisions and extensions)

Student Papers

- Baril, L., Leblanc, S., *Multiplicité de la représentation indicatrice*. Submitted to: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2025) 
Supervisor: [Thomas Brüstle](#)
- Leblanc, S., *Dégénération des représentations de carquois de type A_3 à deux puits*. To appear in: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2024) 
Supervisors: [Emily Cliff](#) and [Shiping Liu](#)
- Leblanc, S., *Transformations de cercles orientés tangents sur la sphère de Riemann*. Submitted to: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2023) 
Supervisor: [Jean-Philippe Burelle](#)

Posters

- Minimal Initial Subcategory*. Presented at Queen's Mathematics Summer School (2026) 
★ This poster won a *Best Poster Award*.
- Multiplicity of the Interval Module*. Presented at AARMS-CMS Student Poster Session (CMS Summer Meeting) (2025) 
Collaborators: Laurianne Baril and Justin Desrochers. Supervisor: [Thomas Brüstle](#).
- Kernel of the Rank Invariant* (with Justin Desrochers). Presented at Summer Research School, Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory (2024) 

Software

knowledgematrix

 GitHub

- A Python library for implementing neural networks and computing their associated *knowledge matrices* (i.e., N_V (Lemma 7.4) in [this paper](#)  and $M(W, f)(x)$ in [this paper](#) .
- Tools: Python

simple_adversarial_detection

 GitHub

- Very simple version of the code used for the experiments in the paper Hidden Activations Are Not Enough: A General Approach to Neural Networks Predictions. [arXiv:2409.13163](https://arxiv.org/abs/2409.13163) 
- Tools: Python

upperhalfplane

 GitHub

- Visualize the action of $\mathrm{PSL}(2, \mathbb{R})$ on the upper half plane (Poincaré half plane model) interactively. samueleblanc.com/software/upperhalfplane 
- Tools: CindyJS, JavaScript, HTML, CSS

riemannsphere

 GitHub

- Visualize the action of $\mathrm{PSL}(2, \mathbb{C})$, i.e., Möbius transformations, and $\mathrm{PSP}(4, \mathbb{R})$ on the Riemann sphere interactively. samueleblanc.com/software/riemannsphere 
- Supervisor: Jean-Philippe Burelle.
- Tools: CindyJS, JavaScript, HTML, CSS

MatTaX

 GitHub

- Chrome Extension and Firefox Add-on that allow the user to convert LaTeX commands into plain text, enabling them to write symbols anywhere. <https://mattalx.org> 
- Tools: JavaScript, HTML, CSS, Bash

Talks

1. *(Co)limit Computation and Its Application to Representation Theory* (January 26, 2026)
Algebra & Geometry Seminar, Queen's University
2. *Generalized Rank in Generalized Linear Algebra* (November 12, 2025)
Grad Seminar, Department of Mathematics and Statistics, Queen's University
3. *Analyse topologique de données* (February 13, 2025)
Club mathématiques de l'Université de Sherbrooke
4. *La propagation avant en tant que matrice* (November 14, 2024) 
Club mathématiques de l'Université de Sherbrooke
5. *Visualisation de transformations sur la sphère de Riemann* (March 21, 2024) 
Club mathématiques de l'Université de Sherbrooke
6. *Théorie des représentations des réseaux de neurones* (October 5, 2023)
Club mathématiques de l'Université de Sherbrooke

Teaching

Guest Lecturer

Real Analysis (MATH/MTHE 328)

Winter 2026

Replaced the course instructor for one week. Delivered 3 hours of lectures.
Department of Mathematics and Statistics, Queen's University

Teaching Assistant

Differential and Integral Calculus (MATH 128)

Winter 2026

Department of Mathematics and Statistics, Queen's University

Méthodes quantitatives en communication marketing (MQG301)

Winter 2025

École de gestion, Université de Sherbrooke

Calcul vectoriel (MAT298)

Fall 2024

Département de mathématiques, Université de Sherbrooke

Statistique appliquée à la gestion (MQG222) <i>École de gestion, Université de Sherbrooke</i>	Summer 2024
Statistique appliquée à la gestion (MQG222) <i>École de gestion, Université de Sherbrooke</i>	Winter 2024

Grader

Differential and Integral Calculus (MATH 121B/128) <i>Department of Mathematics and Statistics, Queen's University</i>	Winter 2026
Real Analysis (MATH/MTHE 328) <i>Department of Mathematics and Statistics, Queen's University</i>	Winter 2026
Algebraic Structures (MTHE 217) <i>Smith Engineering, Queen's University</i>	Fall 2025
Calculus I (APSC 171) <i>Smith Engineering, Queen's University</i>	Fall 2025
Differential and Integral Calculus (MATH 121) <i>Department of Mathematics and Statistics, Queen's University</i>	Fall 2025
Mathématiques discrètes (MAT120) <i>Département de mathématiques, Université de Sherbrooke</i>	Fall 2024

Tutor

Mathematics, 10th grade <i>Volunteering with Le Diplôme avant la Médaille </i>	2023 - 2024
Algèbre linéaire et géométrie vectorielle (MAT902) <i>Université de Sherbrooke</i>	Summer 2023
Biomécanique humaine (KIN325) <i>Université de Sherbrooke</i>	Winter 2023

Service

Academic Coordinator, Queen's GMS During the 2026-2027 academic year, Trevor Shillington and I will be the academic coordinators of the Graduate Mathematics and Statistics Society at Queen's University. We will organize the Graduate Student Seminar of the Department of Mathematics and Statistics.	2026 - 2027
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Selected Academic Activities

Queen's Mathematics Summer School Attended the Queen's Mathematics Summer School at Queen's University (Kingston, ON). June 22 to 26, 2026.	Summer 2026
Auslander Conference Attended the Maurice Auslander Centenary Lectures and International Conference at the Oceanographic Institution (Woods Hole, MA, USA). April 22 to 27, 2026.	Winter 2026
35th RTA Meeting Attended the 35th Meeting on the Representation Theory of Algebras and Related Topics at the Université de Sherbrooke (Sherbrooke, QC). October 23 to 25, 2025.	Fall 2025
Route 81 Conference Attended the Route 81 Conference at Queen's University (Kingston, ON). September 27, 2025.	Fall 2025
Canadian Mathematical Society (CMS) Meeting Attended the 2025 CMS Summer Meeting at the Université Laval (Québec, QC).	Summer 2025

June 7 to 9, 2025.

34th RTA Meeting

Fall 2024

Attended the 34th Meeting on the Representation Theory of Algebras and Related Topics at the Université de Sherbrooke (Sherbrooke, QC). October 4 and 5, 2024.

Research School

Summer 2024

Attended the Summer Research School: Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory, at the UQAM (Montréal, QC). June 3 to 7, 2024.

Introduction to Research (MAT523): Topological Data Analysis

Winter 2024

Optional course. *Département de mathématiques, Université de Sherbrooke*
Supervisor: [Thomas Brüstle](#)

BIRS Workshop

Winter 2024

Assisted (online) to the BIRS Workshop: Representation Theory and Topological Data Analysis. April 8 to 11, 2024.

Research internship: Representation Theory of Quivers

Summer 2023

Département de mathématiques, Université de Sherbrooke
Supervisors: [Emily Cliff](#) and [Shiping Liu](#)

Experimental Mathematics Lab (MAT001): Projective Geometry

Winter 2023

Course taken beyond B.Sc. requirements. *Département de mathématiques, Université de Sherbrooke*
Supervisor: [Jean-Philippe Burelle](#)

Languages

French (native), English (advanced).